

Nightmares Course: Hyperkalemia

Section 1: Case Summary

Scenario Title:	Nightmares Course: Hyperkalemia
Keywords:	Medicine, Cardiology, ACLS
Brief Description of Case:	This case involves the diagnosis and management of hyperkalemia. If not treated appropriately the patient will progress to ventricular fibrillation arrest.

Goals and Objectives	
Educational Goal:	Recognition of hyperkalemic emergency and treatment of hemodynamically unstable bradycardia
Objectives: (Medical and CRM)	1. Review an approach to and differential for bradycardia 2. Provide specific treatment for severe hyperkalemia 3. Prioritize initial investigations and therapies 4. Maintain situation awareness of an unstable patient 5. Communicate the clinical situation clearly to consultants
EPAs Assessed:	TD 1: Recognizing the unstable/critically ill patient, mobilizing the health care team and supervisor, and initiating basic life support.

Learners, Setting and Personnel			
Target Learners:	☒ Medical Students	☒ Junior Residents	☐ Senior Residents
	☐ Staff Physicians	☐ Inter-professional	☐ Other
Location:	☒ Sim Lab	☐ In Situ	☐ Other
Recommended Number of Facilitators:	Instructors: 1		
	Confederates: 1		
	Sim Techs: 1		

Scenario Development	
Date of Development:	2015
Scenario Developer(s):	Dr. Tim Chaplin
Affiliations/Institutions(s):	Queen's University
Contact E-mail:	
Last Revision Date:	Jan 10/2020
Revised By:	Dr. Chris Heyd
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Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: Doug Smith		Age: 67		Gender: Male	
Weight: 80kg					
Presenting complaint: Generally unwell / Medication error					
Temp: 36.5	HR: 40	BP: 90/50	RR: 18	O ₂ Sat: 99%	FiO ₂ : room air
Triage note: You've been called to assess a 67M on the general medical floor. He was admitted 3 days ago for a community acquired pneumonia and is now awaiting discharge home once out-patient services can be put in place. He was noted to be hypokalemic on labs this morning (3.2 mEq/L) and the daytime resident ordered KCl 10mEq in 100cc NS bolus, to be given once. On her initial assessment, the overnight nurse found that he was actually placed on an infusion over the last 10 hours and the patient is now confused and bradycardic.					
Allergies: NKDA					
Past Medical History:			Current Medications:		
HTN			IV KCl replacement		
COPD			Ceftriaxone		
Smoker			Azithromycin		
			Ramipril		
			HCTZ		

Section 2B: Extra Patient Information

A. Further History
Patient vague and repetitive, unable to provide any useful history.
B. Physical Exam
Mental status: mild confusion, repetitiveness Otherwise unremarkable exam

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Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin (<i>Adult</i>)
☐ Standardized Patient
☐ Task Trainer
☐ Hybrid
B. Special Equipment Required
Standard airway equipment Defibrillator
C. Required Medications
ACLS meds Salbutamol nebulizer Insulin/glucose
D. Moulage
IV line in place
E. Monitors at Case Onset
☐ Patient on monitor with vitals displayed
☒ Patient not yet on monitor
F. Patient Reactions and Exam
Confusion

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Section 4: Confederates and Standardized Patients

Confederate and Standardized Patient Roles and Scripts	
Role	Description of role, expected behavior, and key moments to intervene/prompt learners. Include any script required (including conveying patient information if patient is unable)
RN	You are concerned about Doug Smith, a patient you've had for the last 2 shifts. On evening rounds you have found him to be bradycardic and feeling "unwell". You note that he has been receiving IV potassium continuously over the course of the day – there seems to have been a clerical error and he received 10 times the ordered dose. You explain this to the resident on his/her arrival.

Simulation Scenario Template

Section 5: Scenario Progression

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Baseline State Rhythm: Sinus brady HR: 40 BP: 90/60 RR: 18 O ₂ SAT: 99% T: 36.5°C GCS: 13 (E4V4M5)	Confusion Repetitive	<u>Expected Learner Actions</u> ☐ Brief history & physical ☐ Call for help ☐ Ask for recent labs ☐ Place on monitor ☐ Place defib pads ☐ EKG (peaked Ts)	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> <u>Triggers</u> <i>For progression to next state</i> - 3 min → 2. Worsening - All actions complete → 2. Worsening	
2. Worsening HR 30 BP: 70/40 GCS: 10 (E3V2M5)	More lethargic Just moans	<u>Expected Learner Actions</u> ☐ IV Calcium gluconate amps x3 ☐ Insulin regular 10 units IV /2 amps D50W ☐ Salbutamol 20mg nebulized ☐ Consider atropine and dopamine/epinephrine as per ACLS guidelines for unstable bradycardia ☐ EKG (sinusoid)	<u>Modifiers</u> - Atropine → no change - First Ca → HR 35, BP 80/50 <u>Triggers</u> - Second Ca → 4. Stabilization - 6 min → 3. VF arrest	
3. VF arrest Rhythm: VF HR -- BP: --/-- GCS: 3	VSA	<u>Expected Learner Actions</u> ☐ Check pulse ☐ CPR ☐ Defibrillation ☐ IV Calcium	<u>Modifiers</u> <u>Triggers</u> - Ca + Defib → 2. Worsening	RN may need to prompt for status change. Can prompt for Ca if leader has not yet considered it.
4. Stabilization HR 55 BP: 100/60 GCS: 15	Alert Cooperative	<u>Expected Learner Actions</u> ☐ Consider other hyperK treatments such as bicarbonate, diuresis/dialysis	<u>Modifiers</u> <u>Triggers</u>	Team leader should hand over case to MRP or other appropriate consultant.



Simulation Scenario Template

		☐ Repeat EKG ☐ Call most responsible physician ☐ Call nephrology ☐ Consider/discuss disclosure of error	- All actions complete → SMR arrives for handover end of case	
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Simulation Scenario Template

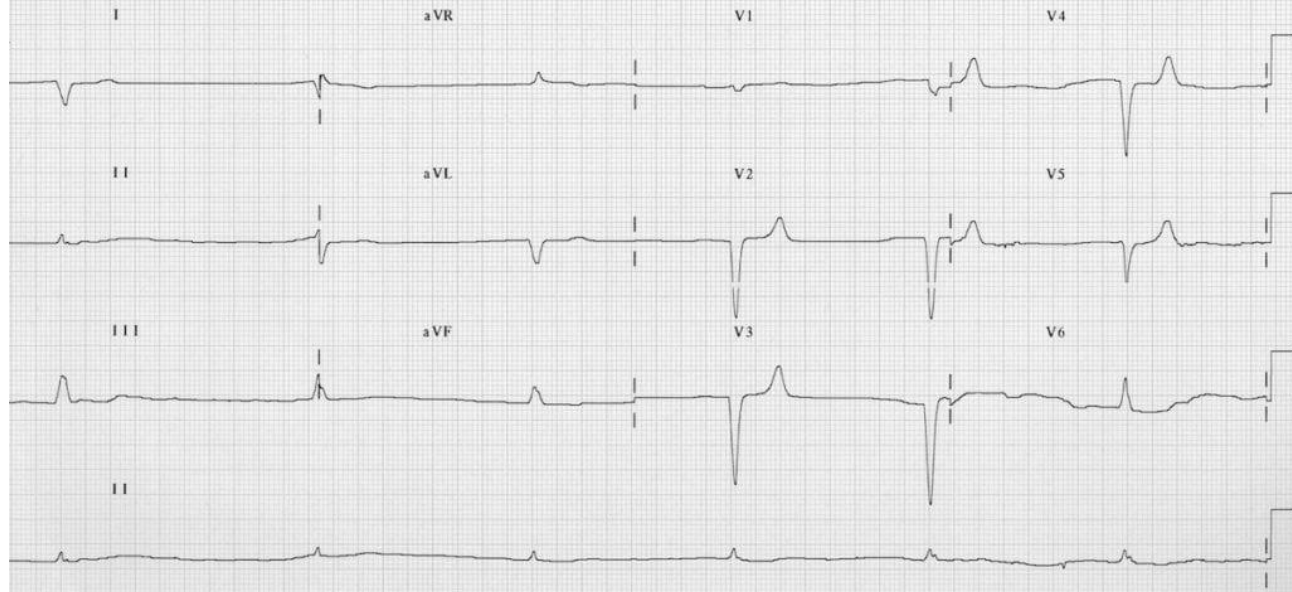
Appendix A: Laboratory Results

<u>CBC</u> WBC 6.1 Hgb 145 Plt 223 <u>Lytes</u> Na 138 K 8.2 Cl 107 HCO ₃ 24 Urea 5.5 Cr 108 Glucose 5.6	
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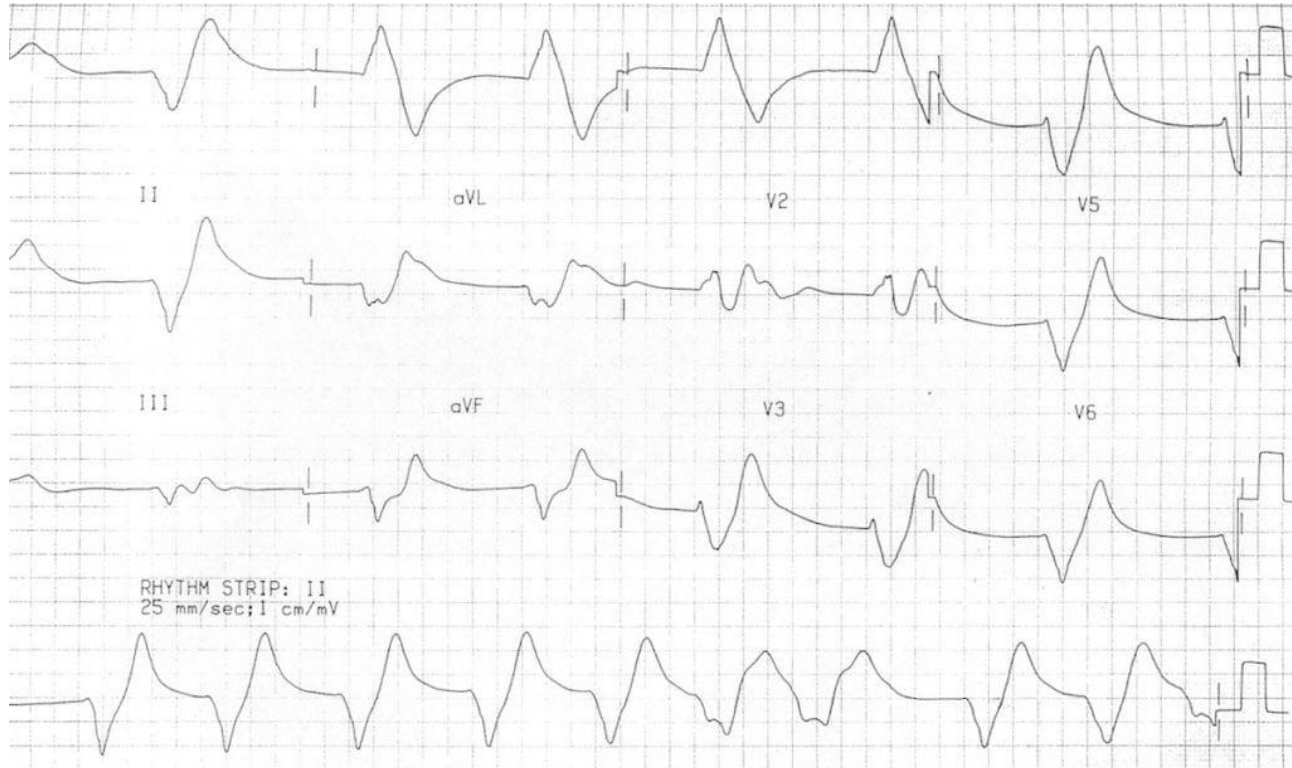
Simulation Scenario Template

Appendix B: ECGs, X-rays, Ultrasounds and Pictures

Peaked Ts



Sinusoid



<https://litfl.com/hyperkalaemia-ecg-library/>

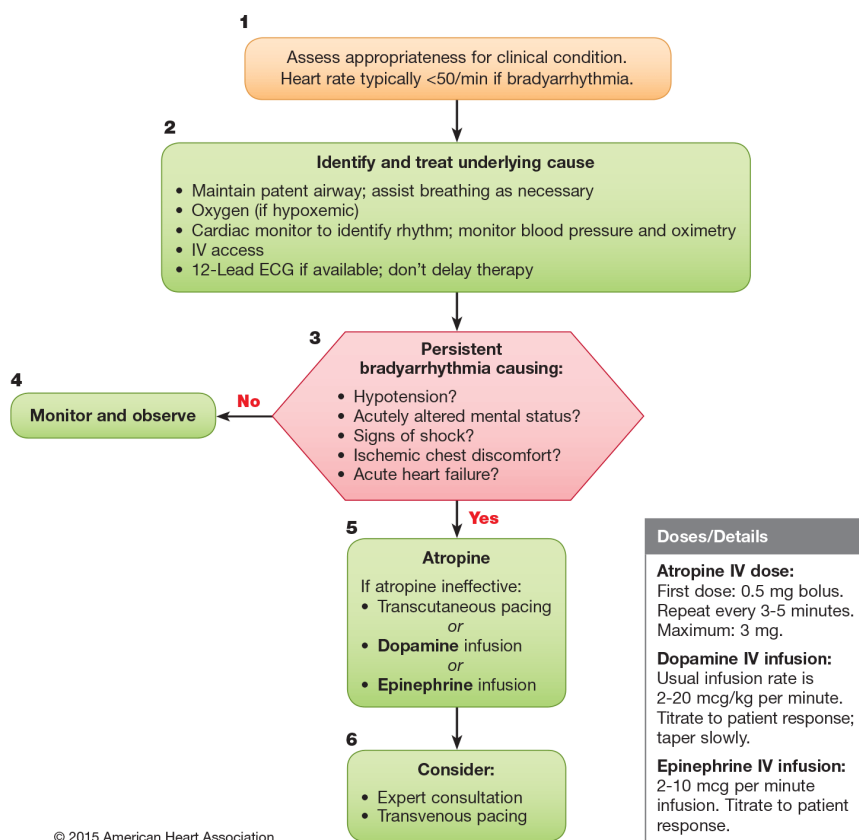
Simulation Scenario Template

Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Include key errors to watch for and common challenges with the case. List issues expected to be part of the debriefing discussion. Supplemental information regarding any relevant pathophysiology, guidelines, or management information that may be reviewed during debriefing should be provided for facilitators to have as a reference.

1. Review the initial approach to bradycardia (ie DIE mnemonic: drugs, ischemia, electrolytes... or any other mnemonic you like!)
2. Discuss the approach to the patient with symptomatic bradycardia
3. Review the medical treatment for severe hyperkalemia
4. Review the options for help during the night (senior resident, RACE team, on-call staff)

Adult Bradycardia With a Pulse Algorithm



References

1. ACLS Bradycardia with a Pulse algorithm
2. <https://emergencymedicinescases.com/emergency-management-hyperkalemia/>
- 3.